EduNet Analyzer

Production Deployment Configurations & Cloud Hosting Guide

Complete Blueprint for Voroa (getvoroa.com), Docker Compose, Render, Railway, Fly.io, and Linux VPS

Target Hosting Platforms:	Voroa (app.getvoroa.com), Docker Compose, Render, Railway, Fly.io
Architecture Type:	React 18 SPA + FastAPI + Native NS-2.35 Discrete-Event Simulator
Deployment Strategies:	Strategy A: Unified Full-Stack Docker (Recommended) Strategy B: Separate Static Site + Python Web Service
Database Engine:	SQLite (WAL Mode) / Managed PostgreSQL via DATABASE_URL
Health Check Route:	GET /api/health (HTTP 200 OK)
Dynamic Port Support:	Auto-binds to $PORT (Defaults to 8000)
WebSocket Streaming:	/api/simulations/ws/{id} and /ws/global/telemetry

EXECUTIVE HOSTING SUMMARY:
Because EduNet Analyzer executes real discrete-event NS-2 (Network Simulator 2) simulations, the application requires the compiled ns and nam Linux binaries. The project provides an optimized multi-stage root Dockerfile that installs NS-2 via apt-get, builds the React frontend with Node 20, and runs FastAPI to serve the API, WebSockets, and frontend assets from a single container. This allows 1-click zero-configuration deployment on Voroa, Render, Railway, and Fly.io without CORS issues or multiple billable services.

1. Voroa Cloud Hosting Configuration (https://app.getvoroa.com)

Voroa is an India-first developer cloud platform offering low-latency deployments in Mumbai. EduNet Analyzer is fully tailored for Voroa's Git-connected deployment flow:

■ Strategy A: Unified Docker Service (Recommended — Full Live NS-2)

Builds both the React SPA and FastAPI backend into a single container containing the native NS-2.35 simulator. Ideal for Voroa's Free and Standard tiers.

Configuration Field	Exact Value to Enter in Voroa	Rationale / Detail
Service Type	**Docker** (Deploy from Dockerfile)	Required for native NS-2 & NAM binaries.
Service Name	edunet-analyzer	Subdomain: edunet-analyzer.getvoroa.com
Git Repository	Devarajb049/EduNet-Analyzer	Connected GitHub repository.
Branch	main	Production release branch.
Root Directory	. (Repository root)	Root contains master Dockerfile.
Dockerfile Path	./Dockerfile	Multi-stage container with Node 20 & Python 3.11.
Health Check Path	/api/health	Verified endpoint returning HTTP 200.
Port	8000	Binds dynamically to $PORT (default 8000).
Auto Deploy	**Yes**	Triggers rebuild on git push.

Strategy B: Two-Service Deployment (Web Service + Static Site)

If you prefer independent services for frontend and backend on Voroa:

Component	Voroa Service Type	Root Dir	Build Command	Start / Output Command
Backend API	Web Service (Python 3.11)	backend	pip install -r requirements.txt	uvicorn app.main:app --host 0.0.0.0 --port $PORT
Frontend UI	Static Site	frontend	npm install && npm run build	Publish Directory: dist

Voroa Environment Variables Specification

Variable	Required	Scope	Value / Purpose
PORT	Auto	Runtime	8000 (Injected by Voroa; server binds to 0.0.0.0).
CORS_ORIGINS	Recommended	Runtime	* or https://edunet-frontend.getvoroa.com
DATABASE_URL	Optional	Runtime	Defaults to SQLite. Supports PostgreSQL: postgresql://...
VITE_API_BASE	Optional	Build-time	/api (default) or backend URL for Strategy B.
VITE_WS_URL	Optional	Build-time	Auto-detected or wss://backend.getvoroa.com/api/simulations

2. Multi-Container Production Docker Compose

For self-hosted Linux VPS (DigitalOcean, AWS EC2, Linode, Hetzner) or on-premise university laboratory servers:

```
version: '3.8'

services:
  backend:
    build:
      context: ./backend
      dockerfile: Dockerfile
    container_name: edunet_backend
    ports:
      - "8000:8000"
    volumes:
      - ./simulation:/app/simulation
      - ./reports:/app/reports
    environment:
      - PYTHONUNBUFFERED=1
      - CORS_ORIGINS=*
      - PORT=8000
    healthcheck:
      test: ["CMD", "curl", "-f", "http://localhost:8000/api/health"]
      interval: 15s
      timeout: 5s
      retries: 3
      start_period: 10s
    restart: unless-stopped

  frontend:
    build:
      context: ./frontend
      dockerfile: Dockerfile
    container_name: edunet_frontend
    ports:
      - "80:80"
      - "5173:80"
    depends_on:
      backend:
        condition: service_healthy
    restart: unless-stopped
```

Docker Deployment Commands:

- **Build and start in background:** docker-compose up -d --build
- **Inspect real-time logs:** docker-compose logs -f backend
- **Verify running services:** docker-compose ps
- **Gracefully stop services:** docker-compose down
- **Access frontend:** http://localhost or http://localhost:5173
- **Access API Swagger docs:** http://localhost:8000/docs

3. Unified Single-Container Master Dockerfile

Used for Voroa Docker Service, Render, Railway, Fly.io, and Cloud Run:

```
# Stage 1: Build React Frontend
FROM node:20-alpine AS frontend-builder
WORKDIR /app/frontend
COPY frontend/package*.json ./
RUN npm ci
COPY frontend/ ./
RUN npm run build

# Stage 2: Production Runtime with NS-2 & FastAPI Backend
FROM python:3.11-slim AS runner
ENV PYTHONDONTWRITEBYTECODE=1 PYTHONUNBUFFERED=1 PORT=8000 HOST=0.0.0.0 \
    STATIC_DIR=/app/frontend/dist SIMULATION_DIR=/app/simulation

WORKDIR /app
RUN apt-get update && apt-get install -y --no-install-recommends \
    build-essential curl ns2 nam && rm -rf /var/lib/apt/lists/*

COPY backend/requirements.txt ./requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

COPY backend/ /app/backend
COPY --from=frontend-builder /app/frontend/dist /app/frontend/dist
RUN mkdir -p /app/simulation/generated /app/simulation/traces /app/reports

EXPOSE 8000
HEALTHCHECK --interval=30s --timeout=5s --start-period=10s --retries=3 \
    CMD curl -f http://localhost:${PORT:-8000}/api/health || exit 1

CMD ["sh", "-c", "python -m uvicorn backend.app.main:app --host 0.0.0.0 --port ${PORT:-8000}"]
```

4. Frontend Nginx Reverse Proxy & WebSocket Configuration

File: frontend/nginx.conf — Configured for SPA client-side routing, static asset caching, and WebSocket connection upgrade:

```
server {
    listen 80;
    server_name localhost;
    root /usr/share/nginx/html;
    index index.html index.htm;

    gzip on;
    gzip_types text/plain text/css application/json application/javascript text/xml image/svg+xml;

    location /assets/ {
        expires 1y;
        add_header Cache-Control "public, max-age=31536000, immutable";
    }

    # REST API Reverse Proxy
    location /api/ {
        proxy_pass http://backend:8000/api/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    # WebSocket Proxy for Real-Time Simulation Streaming
    location /api/simulations/ws/ {
        proxy_pass http://backend:8000/api/simulations/ws/;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_read_timeout 86400s;
        proxy_send_timeout 86400s;
    }

    # SPA Routing Fallback
    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

5. Multi-Cloud Deployment Configurations (Render, Fly.io, PaaS)

Platform	Configuration File	Key Configuration Directives
Render.com	render.yaml	Type: web, env: docker, dockerfilePath: ./Dockerfile, plan: free, healthCheckPath: /api/health
Fly.io	fly.toml	app = 'edunet-analyzer', primary_region = 'bom' (Mumbai), internal_port = 8000, auto_stop_machines = 'stop'
Railway / Heroku	Procfile	web: uvicorn backend.app.main:app --host 0.0.0.0 --port ${PORT:-8000}
Vercel (Frontend)	vercel.json	buildCommand: cd frontend && npm i && npm run build, outputDirectory: frontend/dist, rewrites: /(.*) → /index.html

6. Potential Deployment Pitfalls & Verified Solutions

Technical resolution guide for common production cloud deployment hurdles:

Identified Problem	Severity	Root Cause	Verified Solution
NS-2 binary ('ns') not found in cloud runtime	HIGH	Standard cloud Python buildpacks do not include C++ NS-2 simulation binaries.	Use **Docker deployment (Strategy A)**. The root Dockerfile executes apt-get install -y ns2 nam natively.
CORS blocking API calls across domains	MEDIUM	Separate frontend and backend subdomains without proper origin headers.	Set CORS_ORIGINS=* or https://your-frontend.getvoroa.com in backend environment variables.
SPA 404 error when refreshing /experiments/1	MEDIUM	Web server looks for physical file /experiments/1 on disk.	**Resolved:** main.py implements serve_spa catch-all route; Nginx implements try_files $uri $uri/ /index.html;.
Port binding collision or timeout	MEDIUM	Application hardcoded to port 8000 when platform injects dynamic $PORT.	**Resolved:** main.py reads int(os.getenv('PORT', 8000)) and binds to 0.0.0.0.
SQLite concurrent lock errors	LOW	Simultaneous trace parser write and UI read locking database file.	**Resolved:** database.py enables SQLite PRAGMA journal_mode=WAL on connection.

7. Pre-Flight Production Verification Checklist

- ■ **Backend Test Suite:** All 7 unit/integration tests passed via pytest (Health, Dashboard, Experiments, Compare, CSV, PDF, System).
- ■ **Frontend Compilation:** Production bundle compiled with zero errors via Vite 5.4.21 (2,722 modules transformed).
- ■ **Health Check Endpoint:** Verified responding at GET /api/health with HTTP 200 JSON.
- ■ **WebSocket Endpoint:** Verified responsive at /api/simulations/ws/{id} with polling fallback.
- ■ **Static SPA Routing:** Fallback route configured to serve index.html for all deep links (/simulation/new, /experiments/1, /tcp-udp).
- ■ **Database Bootstrapping:** Initial table creation and benchmark seeding automated during FastAPI lifespan startup.
- ■ **PDF Lab Reports:** In-memory generation verified with ReportLab 4.4.4.
- ■ **Clean Git Working Tree:** .dockerignore configured to exclude node_modules, .git, and cache files.

DEPLOYMENT READY: *EduNet Analyzer is fully validated and ready for 1-click deployment on* ***Voroa (app.getvoroa.com/new/choose)*** *using the Docker runtime. Connect Devarajb049/EduNet-Analyzer, specify ./Dockerfile, set port 8000, and launch your live NS-2 telemetry platform.*